

Project summary: Graph-conditioned state-space models for genetic perturbation response prediction

Aryan Padarathi and Riley Huynh (Allen High School) · Draft for *Bioinformatics* (Oxford) · Code: <https://github.com/The-MathKing/gcpmamba>

Problem. CRISPR Perturb-seq screens measure how the whole transcriptome responds when genes are switched on or off, but only a tiny fraction of gene *combinations* can be measured (131 of 5,460 possible pairs in the Norman et al. 2019 screen). Models that predict unmeasured perturbations—and especially *genetic interactions*, where two perturbations together do something other than the sum of their effects—are a major goal. Recent work (Ahlmann-Eltze et al., *Nat. Methods* 2025) showed that deep models rarely beat simple baselines such as “add up the two single effects”.

Our idea. Mamba-style state-space models (SSMs) process long sequences in linear time, so they can take all ~5,000 measured genes as tokens. Genes have no natural order and an SSM does not know which genes are related, so we *conditioned the SSM’s discretization step on a gene co-expression graph*: genes close (in the graph) to the perturbed gene take larger update steps. We call it GCP-Mamba.

What we did. Benchmarked on the 5 standard GEARS splits of the Norman K562 CRISPRa screen and on the Adamson CRISPRi screen, against 5 simple baselines and GEARS, with 8 ablations (graph removed, permuted, relocated; gene order randomised; recurrence removed; additive prior), 3 random initialisations per split, and Holm-corrected tests.

Results (MSE on the top-20 differentially expressed genes; lower is better).

Model	All test perturbations	Unseen single genes	Doubles, both singles seen
Predict no change	0.561	0.346	0.647
Predict training mean	0.428	0.270	0.485
Additive (sum of single effects)	0.224	0.279	0.048
Linear model (Ahlmann-Eltze 2025)	0.349	0.269	0.296
GCP-Mamba (ours)	0.290	0.261	0.196
Same model, graph removed	0.278	0.265	0.169

- The **additive baseline is the best predictor**; no model explains any genetic-interaction variance beyond additivity.
- The **graph and the graph-conditioned step size give no reliable benefit**: removing or permuting the graph does not make predictions worse. GEARS (Gene Ontology graph; one split, reduced training) *did* predict unseen genes better (0.214), suggesting the co-expression prior is the weak part.
- The scan scales linearly (self-attention runs out of memory at 5,000 genes), but a sparse graph network is 6–25× faster and uses 25–50× less memory.

Contributions beyond the model (evaluation pitfalls).

1. *Input encoding.* Our first version encoded perturbations only over the 500 most variable genes; ~80% of training perturbations silently became all-zero inputs, which invalidated its conclusions.
2. *Seeds.* A single-initialisation comparison showed a “significant” graph effect that vanished with 3 initialisations (initialisation alone moves the error more than the graph does).
3. *Metric.* The genetic-interaction *Pearson* metric used in earlier work ranks models almost in reverse of the interaction variance they explain; we propose a variance-explained measure (R_{GI}^2).

Status and where we would value advice. Full 7-page draft and supplement are written; all numbers regenerate from the repository. Open questions: whether the negative/evaluation framing suits *Bioinformatics* or a different venue; how to strengthen the GEARS comparison (we are CPU-only); whether to swap in a knowledge graph; and whether our statistics are appropriate.